6 **(a)** Fig. 6.1 shows the variation of binding energy per nucleon with the number of nucleons in nucleus.

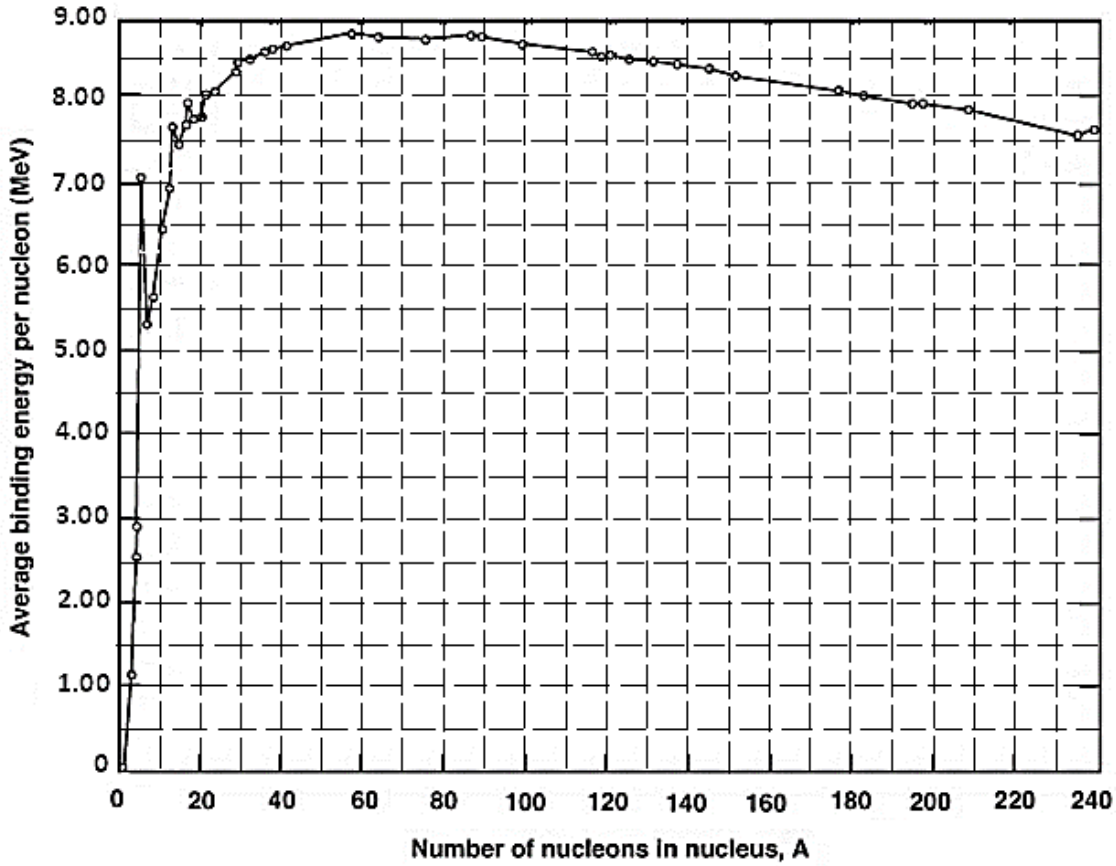

Fig. 6.1

(i) Define binding energy of a nucleus.

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..... [1]

(ii) Using Fig. 6.1, estimate the binding energy of the nucleus Iridium-170, $^{170}_{77}Ir$.

binding energy of $^{170}_{77}Ir$ =..... MeV [1]

(iii) Hence calculate the mass defect of $^{170}_{77}Ir$.

mass defect of $^{170}_{77}Ir$ =..... u [2]

(b) Stellar nucleosynthesis is a collective term for nuclear reactions taking place in stars. These reactions create nuclei of elements heavier than hydrogen.

The "triple" alpha process is a nuclear fusion reaction that occurs in stars where three alpha particles combine to form carbon-12, ${}^{12}_{6}C$.

(i) Determine the energy released in this reaction.

Given:

mass of alpha particle = 4.002603 u

mass of ${}^{12}_{6}C$ = 12.000000 u

energy released =..... J [2]

(ii) "Silicon" burning is the final stage of fusion in massive stars. This process involves silicon-28, ${}^{28}_{14}Si$ capturing multiple alpha particles, until the sequence terminates at ${}^{56}_{28}Ni$. At this point the star can no longer release energy via nuclear fusion. This eventually results in a catastrophic collapse of the star.

Suggest a reason why the star can no longer release energy via nuclear fusion.

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..... [1]